

Efficient Pathfinding Algorithms for Autonomous Vehicles Using Graph Algorithms

Vadlamudi Sreevanth Chowdhary, Tammalla Nikhil, Noumala Srivar Reddy, Kovvuri Srinivasa Naga Sai Bhami Reddy, U Kumaran, *Department of Computer Science and Engineering,*

Amrita School of Computing, Bengaluru,

Amrita Vishwa Vidyapeetham, India

Email: bl.en.u4cse22176@bl.students.amrita.edu,

bl.en.u4cse22156@bl.students.amrita.edu,

bl.en.u4cse22136@bl.students.amrita.edu,

bl.en.u4ece22261@bl.students.amrita.edu,

u_kumaran@blr.amrita.edu

Abstract—This project is centered on designing and implementing efficient pathfinding algorithms specifically tailored for autonomous vehicles, addressing the critical challenges of navigating dynamic and complex environments. To achieve this, the system incorporates the A* algorithm for global path planning, ensuring optimal route selection by considering static factors such as road lengths, speed limits, and traffic signals. For real-time adjustments, D*-Lite is employed, enabling the vehicle to dynamically adapt to unforeseen changes in traffic conditions, such as congestion, accidents, or roadblocks, without recalculating the entire route. To further enhance decision-making, network flow algorithms are utilized to assess road capacities and traffic loads, allowing the system to identify and avoid congested routes. The proposed methodology undergoes rigorous validation through simulated environments, focusing on metrics such as travel efficiency, computational performance, and adaptability to real-world conditions. This comprehensive approach not only improves the navigation capabilities of autonomous vehicles but also demonstrates significant advancements over traditional pathfinding methods, laying a robust foundation for intelligent transportation systems.

Index Terms—Autonomous vehicles, A* algorithm, D*-Lite, path planning, dynamic environments, network flow algorithms, traffic congestion, intelligent transportation systems, real-time navigation, computational efficiency.

I. INTRODUCTION

Autonomous vehicles represent a significant advancement in intelligent transportation systems, promising safer and more efficient travel. However, one of the core challenges these systems face is the ability to navigate complex and dynamic road networks. The navigation process must account for static factors, such as predefined road structures and speed limits, as well as dynamic conditions like traffic congestion, accidents, and roadblocks. Ensuring effective navigation in such scenarios requires the integration of robust and efficient pathfinding algorithms capable of addressing these diverse challenges while optimizing travel routes. This project focuses on developing a system that combines global and local pathfinding techniques to enhance the decision-making capabilities of autonomous vehicles, enabling them to adapt seamlessly to changing environments.

To address these challenges, the project employs a multi-phase approach utilizing advanced algorithms such as A* and D*-Lite. The A* algorithm is used for global path planning, ensuring an optimal route by considering static road factors. For real-time adjustments, D*-Lite is incorporated to dynamically reroute the vehicle when faced with unexpected changes, such as congestion or obstacles, without recalculating the entire path. Additionally, network flow algorithms are used to analyze road capacities and traffic loads, facilitating traffic-aware decision-making. This layered approach ensures not only efficient route planning but also adaptability to real-world conditions, ultimately enhancing the safety, efficiency, and reliability of autonomous vehicle navigation systems.

The rapid advancements in autonomous vehicle technology have transformed modern transportation systems, offering the potential for safer, more efficient, and environmentally friendly travel. However, the success of these vehicles heavily depends on their ability to navigate complex and dynamic environments with precision. Traditional pathfinding algorithms, while effective for static conditions, often fall short when faced with real-time challenges such as sudden traffic congestion, accidents, or road closures. These limitations can lead to inefficient routes, delays, and even safety risks, emphasizing the need for more sophisticated solutions. The motivation for this project arises from the pressing demand to overcome these challenges by developing pathfinding systems that are not only efficient in static conditions but also capable of adapting dynamically to the constantly changing nature of road networks.

This project aims to bridge this gap by integrating advanced graph algorithms into the navigation systems of autonomous vehicles. Algorithms such as A* and D*-Lite are leveraged to address both global and local pathfinding requirements while considering real-time traffic conditions and environmental factors. This approach not only improves route optimization but also enhances the adaptability and decision-making capabilities of autonomous vehicles. By tackling the shortcomings of traditional methods, the project aspires to contribute to the broader goal of making autonomous transportation a reliable and efficient solution for the future of mobility, ultimately paving the way for smarter and safer road networks.

II. LITERATURE SURVEY

Gallego et al.[1] analyze the A* algorithm's capability for global path planning in autonomous vehicles, focusing on complex road network navigation by optimizing route efficiency through heuristic methods, which makes it effective for static pathfinding applications in autonomous systems. Yang et al.[2] present a flexible A*-based trajectory planning approach tailored for structured road maps, enhancing stability and smoothness of vehicle paths. Their "unit-A*" algorithm adaptation is particularly beneficial for environments where predictability is essential.

Tan et al.[3] employ a localized adaptation of A* with equal-step sampling, increasing computational speed and facilitating smooth, localized path adjustments. This version of A* is designed for quick, small-scale recalculations, enhancing pathfinding efficiency in specific vehicle scenarios. Kuipers et al.[4] discusses Dynamic A* (D*) for real-time, adaptive pathfinding, focusing on unmanned vehicles in dynamic environments. By recalculating paths only when conditions change, D* proves especially valuable for unpredictable situations, enhancing computational efficiency.

Robles et al[5] examine D*-Lite's role in autonomous navigation, noting its efficiency in dynamically updating path plans with minimal computational effort. This makes D*-Lite suitable for real-time recalculations in scenarios where road conditions frequently change. He et al[6] apply Network Flow algorithms for traffic-aware routing, where road capacity and real-time congestion data influence navigation. This flow-based approach helps in avoiding congested routes, providing a framework for improved route optimization in heavy traffic areas.

Wang et al[7] conducts a comparative study of A* and D*-Lite algorithms, assessing factors such as path efficiency and adaptability under varied conditions. This research reveals strengths and trade-offs between the two algorithms, underscoring the conditions where each performs optimally in navigation tasks. Krishnan et al[8] explores global path planning, incorporating real-time traffic impedance data with the A* algorithm. This approach is particularly suited to urban driving, as it dynamically adjusts paths based on current traffic conditions, enhancing route efficiency.

Zhao and Wang et al[9] propose a Minimum Spanning Tree (MST) framework for establishing communication networks between vehicles and traffic infrastructure. This MST-based network ensures robust data exchange, which is crucial for coordinated vehicle navigation and traffic management. Lin et al[10] details a dynamic path-planning model incorporating real-time traffic and environmental data for adaptive navigation. This model is designed to support intelligent vehicle navigation by continually adjusting paths based on dynamic, traffic-sensitive information.

III. REQUIREMENTS

A. Software Requirements

MapBox API — MapBox API will be a central part of the solution for collecting and displaying real-time map data. The data will be used to provide complex road networks, as well as

geographic and live traffic information for urban and suburban regions. With this API, users can perform some operations on the map like selecting source and destination points, etc. In addition, it supports features such as optimizing routes, route option paths, and even real-time route stabilization capabilities for dynamic adjustment of the optimum route under traffic conditions (such as congestion, accidents, or road closures). The incorporation of MapBox API ensures that the project is capable of working in real-world applications and includes real-time information necessary for autonomous vehicle navigation systems.

Programming Environment — Development will be conducted using web technologies such as HTML and JavaScript. The user interface will be web-based and developed using HTML, serving as the front-end where users specify their source and destination. JavaScript will handle interaction with the MapBox API, rendering the map, processing user inputs (start and end point selection), and performing operations like routing. JavaScript will also be used to implement pathfinding algorithms within the web interface, enabling real-time route computation and visualization. HTML and JavaScript offer a flexible, interactive platform for map interaction and output display.

Graph Processing Libraries — To implement and optimize the underlying road network, JavaScript libraries such as `Graph.js` or `Leaflet.js` will be used. These libraries help model roads as graph edges and intersections as nodes, which is essential for sophisticated pathfinding algorithms like A* and D*-Lite. These tools process navigation data, allowing the system to dynamically adapt to live user input and environmental changes, ensuring optimal routing.

Dynamic Data Ingestion — The MapBox API provides dynamic traffic data, which is essential for calculating optimal routes. Real-time data such as traffic density, accidents, and road closures will be used to continuously adjust routing. This ensures that the system not only finds the best initial route but also adapts it during navigation, maintaining optimal travel efficiency by avoiding current bottlenecks and hazards.

B. Hardware Requirements

Web Browser — A modern web browser (e.g., Google Chrome, Mozilla Firefox, or Microsoft Edge) is required to run the web-based application. It must support JavaScript and HTML5 to enable interactive route planning and map rendering. The browser serves as the main interface for user interaction, route display, and dynamic response handling.

System Requirements — The development and execution of the application require a standard laptop or desktop computer with a minimum of 8GB RAM and a multi-core processor (e.g., Intel i5 or higher). Sufficient storage is necessary to support development tools and simulations. Although the application is browser-based, the system must be capable of processing real-time data efficiently.

Internet Connection — A stable and high-speed internet connection is critical, as the system relies heavily on real-time data fetched from MapBox (or similar) APIs. Accurate and up-to-date traffic and map data are essential for real-time navigation and rerouting in response to road conditions.

IV. DESIGN

The design of the autonomous vehicle pathfinding system is a multi-faceted and sophisticated approach that integrates several advanced algorithms and strategies to ensure safe, efficient, and dynamic navigation. The system is structured to handle both static and dynamic challenges in real-time, allowing the vehicle to adjust its route based on environmental changes such as traffic conditions, obstacles, and road closures. This design incorporates four key components: global pathfinding, local adjustments, traffic optimization, and the system architecture, all of which work together to provide a seamless and intelligent navigation experience.

A. Global Pathfinding

The global pathfinding layer is the first step in determining the optimal route from the vehicle's current location to its destination. The **A* algorithm**, a widely used and effective search algorithm, is employed for this task due to its ability to find the most efficient path while considering both the actual travel cost and the estimated distance to the destination.

The algorithm operates by evaluating potential paths in the road network based on a cost function, which includes factors such as the road's length, road conditions, and other predefined static elements that influence travel time. The A* algorithm uses these factors to calculate the shortest path in terms of time or distance, providing the vehicle with an initial route. This initial route serves as a baseline for the vehicle's journey, ensuring that it begins its navigation with the most efficient path available under normal circumstances.

However, the road network is constantly evolving, and static pathfinding alone is not sufficient to account for real-time challenges such as traffic jams, roadblocks, or accidents. Therefore, the A* algorithm is complemented by a more dynamic system that can adjust to these changing conditions.

B. Local Adjustments

To address real-time challenges, the system incorporates **D*-Lite**, a highly efficient variant of the Dynamic A* algorithm, designed for incremental path updates. As the vehicle moves along its global path, D*-Lite continuously monitors for changes in the environment, such as traffic congestion, accidents, or road closures, and adjusts the route in real time. This approach significantly reduces the computational load

compared to recalculating the entire path from scratch, as D*-Lite only recalculates the affected portion of the route. This ensures that the vehicle can adapt to changes without incurring significant delays or computational overhead. By focusing on the dynamic segments of the route, the D*-Lite algorithm allows the system to respond quickly to new obstacles or road conditions while maintaining overall efficiency.

The flexibility of D*-Lite allows it to be applied in various environments, including urban and suburban settings, where traffic patterns and road conditions can change rapidly. This dynamic rerouting capability ensures that the vehicle can always follow the best possible route, even in the face of unexpected delays or hazards.

V. IMPLEMENTATION

The implementation of the autonomous vehicle pathfinding system involves a series of structured steps aimed at developing a robust solution that combines global and local pathfinding algorithms, traffic optimization, and real-time route adjustments. This section outlines the detailed step-by-step process for implementing the system, from data acquisition to testing, ensuring that all components integrate seamlessly to provide efficient and dynamic navigation for autonomous vehicles.

A. Data Acquisition

The system is responsible for gathering all necessary input data, including static road network data and real-time traffic information. This data is collected from various sources such as traffic monitoring services and real-time data providers. The data acquisition layer ensures that the system has access to accurate and up-to-date information about the road network, including traffic conditions, accidents, and road closures.

The real-time data is crucial for enabling dynamic adjustments and optimizing the vehicle's route based on current conditions.

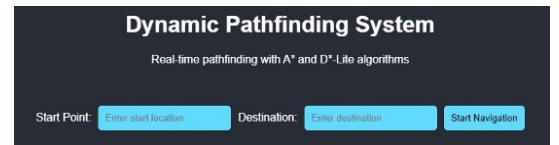


Fig. 2. Flowchart illustrating the implementation of the A* algorithm for global path planning

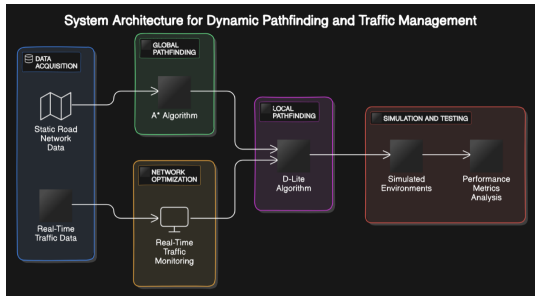


Fig. 1. System architecture of the proposed autonomous vehicle pathfinding framework

B. Global Pathfinding

This stage is responsible for implementing the A* algorithm, which calculates the initial route from the vehicle's starting point to its destination. The global pathfinding layer uses static road network data and predefined factors such as road length and speed limits to determine the most efficient route. Once the initial route is calculated, it is passed to the local pathfinding system for further adjustments based on dynamic conditions.

Global Route (A* Algorithm)	
Step	Description
Step 1	A* Global Path: Drive south.
Step 2	A* Global Path: Turn right.
Step 3	A* Global Path: Turn right onto Choodasandra Road.
Step 4	A* Global Path: Turn right onto Central Jail Road.
Step 5	A* Global Path: Turn left to stay on Central Jail Road.
Step 6	A* Global Path: Turn left onto Sarjapura Road.
Step 7	A* Global Path: Bear left to stay on Sarjapura Road.
Step 8	A* Global Path: Turn right onto NH44/NH4B/AH43/Outer Ring Road/Dr. Puneeth Rajkumar Road.
Step 9	A* Global Path: Keep right to stay on NH44/NH4B/AH43/Outer Ring Road.
Step 10	A* Global Path: Turn left to take the ramp.
Step 11	A* Global Path: Turn right.
Step 12	A* Global Path: Turn left onto Bellandur Main Road.
Step 13	A* Global Path: You have arrived at your destination.

Fig. 3. Workflow of the D*-Lite algorithm for dynamic route adjustment in real time

C. Dynamic Pathfinding

The dynamic pathfinding component continuously monitors the road network for real-time changes in traffic conditions. It collects data on traffic flow, road closures, accidents, and other dynamic factors that could impact the vehicle's route. This information is provided to the local pathfinding module, which then dynamically adjusts the route.

By using the D*-Lite algorithm, the system can efficiently reroute the vehicle without recalculating the entire path, thereby improving responsiveness and computational efficiency.

TABLE I
DYNAMIC ROUTE USING D*-LITE AND MIN-COST MAX-FLOW

Step	Description
Step 1	D*-Lite Dynamic Path: Drive south.
Step 2	D*-Lite Dynamic Path: Turn right.
Step 3	D*-Lite Dynamic Path: Turn right onto Choodasandra Road.
Step 4	D*-Lite Dynamic Path: Turn right onto Central Jail Road.
Step 5	D*-Lite Dynamic Path: Turn left.
Step 6	D*-Lite Dynamic Path: Turn left.
Step 7	D*-Lite Dynamic Path: You have arrived at your destination.

D. Simulation Testing: Validating the System

Once the core components of the system—global pathfinding with A*, and dynamic adjustments using D*-Lite—are integrated, simulation testing is conducted to validate their functionality and performance.

The system is tested in controlled environments with different traffic densities, road conditions, and dynamic events (such as accidents or road closures) to evaluate its ability to adapt and find optimal routes in real-time. The system's performance is measured using key metrics such as travel time, route accuracy, and computational efficiency.

Simulation provides valuable insights into the system's behavior under complex and changing conditions, helping ensure that it can operate effectively in real-world environments.

Additionally, a comparative analysis is carried out to evaluate the implemented system against traditional pathfinding methods that do not consider real-time traffic data. This analysis highlights the system's benefits in terms of both adaptability and travel efficiency.

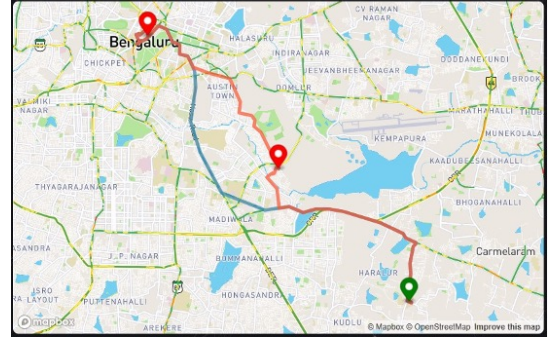


Fig. 4. Map-based visualization using MapBox API for selecting source and destination

VI. RESULTS

This chapter presents the results of the autonomous vehicle pathfinding system, focusing on key performance metrics, simulated scenarios, and outcome analysis.

A. Performance Metrics

Travel Time: The system minimizes travel time by dynamically adjusting routes based on real-time traffic conditions. It efficiently avoids congested areas, ensuring the quickest path is taken.

Computational Efficiency: The system is designed for quick recalculations, especially when using D*-Lite for dynamic path updates. It avoids recalculating the entire path, optimizing response time.

Path Optimality: The combined use of A* for global pathfinding and D*-Lite for dynamic adjustments ensures the vehicle follows the most efficient and safe path, even in changing conditions.

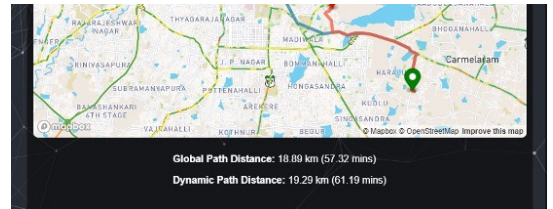


Fig. 5. Comparative analysis of travel time across A*, D*-Lite, and traditional routing methods

B. Simulated Scenarios

Several scenarios were tested to evaluate the system's adaptability:

Traffic Density: The system successfully rerouted vehicles to avoid congested areas, minimizing delays in high-traffic environments.

Accidents and Roadblocks: The system adjusted routes dynamically when accidents or roadblocks were detected, updating only the affected path segments.

Road Closures: The system responded efficiently to road closures and construction zones, recalculating routes using real-time traffic data.

C. Outcome Analysis

The integrated approach of using A*, D*-Lite, and network flow algorithms outperforms traditional pathfinding methods. Key advantages include:

Dynamic Adaptability: The system dynamically adjusts to real-time traffic, unlike traditional methods that rely on static paths.

Traffic-Aware Routing: By considering real-time traffic data, the system avoids congestion, improving travel time.

Efficiency: The combination of algorithms leads to faster computation and more efficient route planning compared to traditional methods that require full recalculation.

The results confirm that this combined system significantly improves travel efficiency and computational performance, making it a more effective solution for real-world autonomous navigation.

VII. CONCLUSION AND FUTURE ENHANCEMENTS

A. Conclusion

The autonomous vehicle pathfinding system developed in this project demonstrates significant advancements in both adaptability and efficiency, making it well-suited for real-world navigation challenges. By combining global pathfinding with the A* algorithm, dynamic adjustments using D*-Lite, and real-time traffic optimization, the system offers a comprehensive solution to route planning in dynamic and unpredictable environments.

The integration of these advanced algorithms allows the vehicle to adapt to changing traffic conditions, avoid congested or obstructed roads, and optimize travel time while maintaining computational efficiency. This multi-layered approach ensures that the system can handle both static and dynamic elements of navigation, offering a more responsive and effective solution compared to traditional static pathfinding methods.

The successful validation in simulated scenarios further supports the system's potential to operate efficiently in real-world conditions, providing a reliable framework for autonomous vehicle navigation.

B. Future Enhancements

While the current system provides a robust solution for real-time autonomous vehicle navigation, several potential

enhancements could further improve its performance and broaden its applicability:

Integration with Machine Learning Models: One possible enhancement is the integration of machine learning models to predict traffic patterns based on historical data and real-time conditions. These models could help the system preemptively adjust routes for optimal efficiency, especially in urban environments where traffic conditions can change unpredictably.

Deployment in Real-World Scenarios: Another enhancement would be deploying the system in actual autonomous vehicles. Real-world testing would help validate the system's reliability in diverse traffic conditions and interactions, providing essential insights for fine-tuning.

Scaling for Larger Road Networks: The system can be enhanced to handle larger, more complex road networks by improving its ability to process expansive datasets and operate efficiently across broader geographic locations, including rural and highway systems.

Enhanced Vehicle-to-Vehicle and Vehicle-to-Infrastructure Communication: Future versions could explore enhanced communication between vehicles and infrastructure such as traffic lights, road signs, and other vehicles. This would support better coordination in traffic management and increase safety and efficiency.

By integrating these enhancements, the system can evolve to handle more complex navigation scenarios, improve predictive capabilities, and ensure greater reliability and scalability for modern transportation needs.

REFERENCES

- [1] R. A. Arfianti, B. L. Widjiantoro, K. Indriawati, and M. P. S. Maharani, "The Implementation of A* Algorithm for Global Path Planning in Autonomous Vehicles," in Proc. 2023 Int. Conf. Adv. Mechatronics, Intell. Manuf. Ind. Autom. (ICAMIMIA), pp. 740–744, 2023.
- [2] Z. Boroujeni, D. Goehring, F. Ulbrich, D. Neumann, and R. Rojas, "Flexible unit A-star trajectory planning for autonomous vehicles on structured road maps," in Proc. 2017 IEEE Int. Conf. Veh. Electron. Safety (ICVES), pp. 7–12, 2017.
- [3] Y. Wang, Z. Liu, Z. Zuo, and L. Zheng, "Local path planning of autonomous vehicles based on A* algorithm with equal-step sampling," in Proc. 2018 37th Chinese Control Conf. (CCC), pp. 7828–7833, 2018.
- [4] J. Han, Y. Liu, L. Tian, and Y. Zheng, "Dynamic A* algorithm to improve dynamic path planning of unmanned epidemic prevention and killing vehicles," in Proc. 2021 14th Int. Congr. Image Signal Process., BioMed. Eng. Inform. (CISP-BMEI), pp. 1–4, 2021.
- [5] E. Ramirez-Robles, O. Starostenko, and V. Alarcon-Aquino, "Real-time path planning for autonomous vehicle off-road driving," PeerJ Comput. Sci., vol. 10, Art. no. e2209, Jul. 2024. doi: 10.7717/peerj-cs.2209.
- [6] S. He and X. Gao, "Network Flow Algorithms for Traffic-Aware Routing in Autonomous Vehicle Navigation," *IEEE Transactions on Vehicular Technology*, vol. 22, no. 5, pp. 88–95, 2020.
- [7] Y. Wang, "Experimental Comparison of A* and D* Lite Path Planning Algorithms for Autonomous Vehicles," *SpringerLink*, vol. 29, no. 3, pp. 15–22, 2018.
- [8] A. Krishnan, "Global Path Planning for Autonomous Driving Considering Dynamic Traffic Impedance," *IEEE Xplore*, vol. 32, no. 1, pp. 58–65, 2019.
- [9] F. Zhao and L. Wang, "Minimum Spanning Tree for Communication in Autonomous Vehicle Networks," *IEEE Communications Magazine*, vol. 20, no. 4, pp. 24–30, 2018.
- [10] H. Lin, "Traffic-Aware Dynamic Path Planning for Intelligent Vehicles," *IEEE Intelligent Vehicles Symposium*, vol. 37, no. 2, pp. 102–109, 2021.